ZAALIMA DEVELOPMENT · DATA ANALYTICS INTERNSHIP

Week 1 • Day 1

Environment Setup & Project Initialization

Installed all tools, created GitHub repository, and built the complete project folder structure

DATE	INTERN NAME	PROJECT	MENTOR
Day 1 of 28 | Week 1	______________________ _____	Customer Churn Prediction & LTV Engine	______________________ _____

■ Technologies Used Today

Python 3.11	VS Code	Git	GitHub	PostgreSQL	pgAdmin 4	pip

■ Today's Key Metrics

9	7,043	21	1
Tools Installed	Dataset Rows	Dataset Columns	GitHub Commits

■ Day Objectives

- ■ Install Python 3.11, VS Code, PostgreSQL 15, pgAdmin 4
- ■ Create GitHub account and project repository
- ■ Clone repository to local machine
- ■ Build complete project folder structure
- ■ Download Telco Customer Churn dataset from Kaggle
- ■ Create requirements.txt and install all Python libraries
- ■ Push first commit to GitHub

■ Tasks Completed Today

1. Installed Python 3.11 with PATH option enabled — verified with 'python --version'

2. Installed VS Code and configured extensions: Python, Pylance, GitLens, PostgreSQL

3. Installed Git and configured global username and email settings

4. Created GitHub account and new public repository: customer-churn-ltv-engine

5. Cloned the repository to local machine using 'git clone'

6. Created all project folders: data/raw, data/processed, notebooks, src/db, src/features, src/models, src/api, models, reports/figures, tests

7. Created all __init__.py files in src/ subdirectories

8. Downloaded Telco Customer Churn Dataset from Kaggle and saved to data/raw/telco_churn.csv

9. Created requirements.txt listing all 14 Python libraries needed for the project

10. Ran 'pip install -r requirements.txt' and confirmed all libraries installed successfully

11. Created README.md with project description, tech stack, and setup instructions

12. Created .env file with PostgreSQL credentials (not committed to Git)

13. Added .env to .gitignore to protect sensitive credentials

14. Ran 'git add . → git commit → git push' for the first time

15. Verified commit appears on GitHub repository page

■ Concepts Learned

- Version control: What Git is and why we track code history
- Repository, Commit, Push, Clone — the four core Git operations
- Why Python must be added to PATH during installation
- What a virtual environment is (for future use)
- Why .env files must never be committed to GitHub
- What requirements.txt does and how pip uses it
- Project folder structure best practices for data science projects
- The difference between local (machine) and remote (GitHub) repositories

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Python 3.11	Latest stable release	✓ Installed
VS Code	Latest	✓ Installed
Git	Latest	✓ Installed
pgAdmin 4	Latest	✓ Installed
PostgreSQL 15	15.x	✓ Installed
Pandas	2.1.4	✓ via pip
Scikit-Learn	1.3.2	✓ via pip

XGBoost	2.0.3	✓ via pip
FastAPI	0.108.0	✓ via pip

■ Outputs / Deliverables Produced

- ❖ GitHub repository: customer-churn-ltv-engine (public, accessible online)
- ❖ Complete local project folder structure (10 directories, 5 __init__.py files)
- ❖ requirements.txt — lists all 14 project dependencies
- ❖ README.md — professional project description
- ❖ .env file — local database credentials (not on GitHub)
- ❖ .gitignore — protects .env and cache files
- ❖ data/raw/telco_churn.csv — 7,043 rows, 21 columns (Kaggle dataset)
- ❖ First GitHub commit: 'Week1-Day1: Setup project structure and environment'

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
Python not recognized after installation	Uninstalled and reinstalled Python with 'Add to PATH' checkbox ticked on first screen
GitHub authentication failed during 'git push'	Created a Personal Access Token on GitHub Settings → Developer Settings and used it as password

■ GitHub Commit Record

Commit Message	Week1-Day1: Setup project structure, installed dependencies, added README and .gitignore
Branch	main
Files Changed	README.md, requirements.txt, .gitignore, src/__init__.py files, folder structure
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ python --version shows 3.11.x	■ git --version shows version number
■ VS Code extensions installed	■ pgAdmin connects to PostgreSQL
■ GitHub repo visible online	■ All project folders created

■ telco_churn.csv in data/raw/	■ Libraries installed via pip
■ .env created (not on GitHub)	■ Day 1 commit on GitHub

■ Self Assessment

8/10	7	Week 1, Day 1	1/28
Self Rating	Hours Spent	Progress	Day in Program

■ Key Learnings & Reflections

★ Git workflow: always 'git add → commit → push' in order every day
★ PATH is critical for Python — always tick the option during installation
★ A clean folder structure from Day 1 saves hours of confusion later
★ The .env file is how we keep secrets out of GitHub — never skip this

■ Plan for Tomorrow

→ Learn PostgreSQL basics: CREATE DATABASE, CREATE TABLE, SELECT
→ Create churn_db database in pgAdmin
→ Import telco_churn.csv into PostgreSQL using Python
→ Verify all 7,043 rows loaded correctly

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED
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